

Pasteurellosis — Comprehensive Disease Characteristics Report

Disease: Pasteurellosis **MONDO ID:** MONDO:0005901 **Category:** Infectious Disease (zoonosis)
Evidence base: 11 confirmed findings, 32 papers reviewed across 5 investigation iterations

Summary

Pasteurellosis is a **zoonotic bacterial infection** caused by Gram-negative coccobacilli of the genus *Pasteurella*, overwhelmingly *Pasteurella multocida*, which are commensals of the oral and respiratory flora of cats, dogs, and many other animals. In humans it is acquired chiefly through **animal bites, scratches, licking of broken skin, or inhalation/aspiration of contaminated animal secretions**. Cats are the source in 60–80% of human cases and dogs in most of the remainder ([PMID: 6371440](#)). Because the etiology is infectious, there is **no human genetic cause** — susceptibility to severe disease is instead governed by host comorbidities (cirrhosis, immunosuppression, chronic lung disease, extremes of age) and by **bacterial virulence factors** (capsule, lipopolysaccharide, and *Pasteurella multocida* toxin).

Human pasteurellosis presents in **three clinical groups**: (1) rapidly progressive soft-tissue infection at a bite/scratch site — the most common presentation, with erythema, warmth, tenderness, and purulent drainage appearing within hours; (2) respiratory-tract infection, predominantly in elderly patients with underlying COPD, bronchiectasis, or malignancy; and (3) invasive/systemic disease (bacteremia, meningitis, endocarditis, septic arthritis, osteomyelitis) concentrated in neonates, the immunocompromised, and cirrhotic hosts ([PMID: 6371440](#); [PMID: 9097378](#); [PMID: 26356688](#)). Notably, **non-bite-associated infection carries a substantially higher risk of bacteremia, ICU admission, and death** than bite-associated infection.

P. multocida is also a **major veterinary pathogen** of global economic importance, causing hemorrhagic septicemia (serotype B:2) in cattle and buffalo with near-100% peracute mortality, fowl cholera in poultry, and progressive atrophic rhinitis in swine (driven by the toxin PMT). The organism's virulence rests on a polysaccharide **capsule** (anti-phagocytic, complement-resistant) and **lipopolysaccharide** (essential for host survival and the antigenic

basis of 16 serovars), both proven necessary for virulence in animal challenge models. First-line human therapy is **penicillin**, with amoxicillin-clavulanate as the practical choice for polymicrobial bite wounds and doxycycline as an alternative. There is no human vaccine; prevention relies on wound care, selective antibiotic prophylaxis, and rabies post-exposure prophylaxis for animal bites.

Section 1 — Disease Information

Overview. Pasteurellosis is an acute zoonotic infection caused by *Pasteurella* species — small, nonmotile, facultatively anaerobic, oxidase-positive, Gram-negative coccobacilli. *P. multocida* is the principal human pathogen; *P. canis*, *P. dagmatis*, and *P. stomatis* are less common. The organism is part of the normal oropharyngeal and gastrointestinal flora of domestic and wild animals; human infection is almost always animal-associated. *P. multocida* holds a historic place in microbiology as the first pathogen shown by Louis Pasteur (1881) to cause fowl cholera and used in his foundational attenuation/vaccination experiments ([PMID: 17107417](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0005901
MeSH	Pasteurella Infections (D010326)
ICD-10	A28.0 (Pasteurellosis)
ICD-11	Zoonotic bacterial disease category (1B/1C area)
NCBI Taxonomy (pathogen)	<i>Pasteurella multocida</i> — txid747
OMIM	Not applicable (no Mendelian human entry — infectious disease)
Orphanet	Not a listed rare genetic disease

Synonyms / alternative names. Pasteurella infection; *Pasteurella multocida* infection. Related animal-specific names: hemorrhagic septicemia (cattle/buffalo), fowl cholera (poultry), snuffles (rabbits), progressive atrophic rhinitis (swine).

Source of information. The human clinical knowledge base derives primarily from **aggregated disease-level resources** — case reports, case series, and retrospective cohort reviews — rather than large EHR-derived cohorts, reflecting the relative rarity and sporadic nature of severe human pasteurellosis. Veterinary and mechanistic data derive from experimental animal models and molecular microbiology.

Section 2 — Etiology

Primary cause. Pasteurellosis is an **infectious disease**; the causal agent is *Pasteurella multocida* (and less often other *Pasteurella* spp.). There is **no genetic (Mendelian) cause** in humans. The primary transmission route is direct inoculation from an animal reservoir — bites, scratches, or licking of broken skin — with a secondary respiratory route via inhalation/aspiration of contaminated secretions ([PMID: 40063964](#); [PMID: 11242756](#)).

Risk factors.

- *Environmental / exposure:* Contact with cats and dogs is the dominant exposure. In neonatal pasteurellosis, cat/dog exposure was the major risk (non-traumatic 44%, traumatic 8%), with **vertical transmission** in 44% ([PMID: 19208671](#)). In infantile meningitis, 61% arose from **indirect** household animal contact ([PMID: 40063964](#)).
- *Host comorbidity:* Cirrhosis/liver dysfunction predisposes to opportunistic bacteremia ([PMID: 7560209](#)); chronic lung disease (COPD, bronchiectasis, malignancy) predisposes to pulmonary infection ([PMID: 9097378](#)); immunocompromise (including HIV) predisposes to invasive disease. A Charlson comorbidity index ≥ 1 was significantly associated with non-bite (more invasive) infection ([PMID: 26356688](#)).
- *Age:* Neonates/infants and the elderly are at highest risk of severe disease.

Genetic risk factors. No human causal variants, susceptibility loci, or modifier genes are established — this is not a heritable disease.

Protective factors. No genetic protective variants apply. Environmental protection is behavioral: prompt wound irrigation, hand hygiene after animal contact, and avoidance of high-risk animal contact by immunocompromised/cirrhotic individuals.

Gene–environment interactions. Not applicable for the human host. The operative "genotype–environment" axis is between **bacterial virulence genotype** (capsular serotype, LPS glycoform, PMT carriage) and **host immune status**.

Section 3 — Phenotypes

Because pasteurellosis is infectious, phenotypes are clinical signs/symptoms rather than heritable HPO traits, but HPO terms are suggested for knowledge-base mapping.

Phenotype	Type	Onset / course	Frequency	Suggested HPO
Cellulitis / soft-tissue infection at bite site (erythema, warmth, tenderness, purulent drainage)	Clinical sign	Acute, rapid (hours)	Most common presentation	HP:0100658 (Cellulitis)
Purulent wound drainage	Clinical sign	Acute	Frequent in local infection	HP:0031357
Septic arthritis (proximal to bite)	Clinical sign	Acute–subacute	Serious local complication	HP:0100776
Osteomyelitis	Clinical sign	Subacute	Local complication, esp. finger/hand after cat bite	HP:0002754
Tenosynovitis / abscess	Clinical sign	Acute–subacute	Local complication	—
Pneumonia / tracheobronchitis / lung abscess / empyema	Clinical sign	Subacute, indolent	2nd most common site	HP:0002090 (Pneumonia)
Bacteremia / sepsis	Laboratory / clinical	Acute, can be fulminant	51% of infantile invasive cases; 37% of non-bite adult cases	HP:0031864 (Sepsis)
Meningitis (± seizures)	Clinical sign	Acute	Seizures in 20% of infantile meningitis	HP:0001287; HP:0001250 (Seizure)
Endocarditis	Clinical sign	Subacute	Rare, severe	HP:0031617

Characteristics. Onset spans **neonatal to geriatric**. Severity ranges from mild self-limited cellulitis to fulminant fatal sepsis (Waterhouse–Friderichsen syndrome reported in cirrhosis, [PMID: 7560209](#)). Local infections are characterized by the **rapid appearance of erythema, warmth, tenderness, and frequently purulent drainage** ([PMID: 6371440](#)). Progression is typically acute; untreated deep infection extends to septic arthritis and osteomyelitis.

Quality-of-life impact. Hand/finger infections after cat bites can cause functional impairment and, if complicated by tenosynovitis or osteomyelitis, prolonged disability. Invasive disease carries substantial mortality (Section 11). Formal EQ-5D/SF-36 data specific to pasteurellosis are not available.

Section 4 — Genetic / Molecular Information

This section is largely NOT APPLICABLE to the human host. Pasteurellosis is not a genetic disease; there are no human causal genes, pathogenic variants, ACMG classifications, allele frequencies, modifier genes, epigenetic signatures, or chromosomal abnormalities.

The relevant molecular genetics belongs to the **pathogen**:

- **Capsule biosynthetic (*cap*) locus** — 15 genes in three functional regions (export regions 1 and 3; biosynthesis region 2). Disruption of the export gene *cexA* yields an acapsular, avirulent mutant ([PMID: 10816499](#)).
- **LPS biosynthesis genes** — heptosyltransferases *hptA/hptB*, Kdo kinase *kdkA*, and outer-core transferases *pcgD* and *hptE*, which determine LPS glycoform and virulence ([PMID: 17517879](#); [PMID: 33663572](#)).
- **toxA** — encodes the 146-kDa *Pasteurella multocida* toxin (PMT), a deamidase virulence factor (Section 6).

Section 5 — Environmental Information

Infectious agent. *Pasteurella multocida* (NCBI txid747) is the principal agent, with subspecies *multocida*, *septica*, and *gallicida*. Other species: *P. canis*, *P. dagmatis*, *P. stomatis*. Capsular serogroups A, B, D, E, F and 16 LPS-based serovars define strain diversity.

Environmental / exposure factors. The reservoir is the animal oral/respiratory tract. Human exposure occurs through the domestic environment (pet ownership), occupational settings (veterinarians, farmers, abattoir workers, animal handlers), and recreational animal contact. There is **no toxin, radiation, or pollution etiology**.

Lifestyle factors. Pet keeping and close animal contact (being licked, sleeping with pets) are the operative behavioral factors, including the **indirect** contact responsible for 61% of infantile meningitis cases (PMID: 40063964).

Section 6 — Mechanism / Pathophysiology

Causal chain (human bite-associated soft-tissue infection)

1. Animal bite/scratch/lick **inoculates** *P. multocida* from animal oral flora into human subcutaneous tissue →
2. Bacterial **capsule** prevents phagocytosis and confers complement resistance, and complete **LPS** enables survival in host tissue → *leads to* local bacterial proliferation →
3. LPS (endotoxin) and bacterial products **trigger innate immune activation** (neutrophil recruitment, cytokine release) → *results in* the characteristic rapid cellulitis (erythema, warmth, tenderness, purulent drainage) within hours →
4. If uncontrolled, contiguous spread → *leads to* abscess, tenosynovitis, septic arthritis, osteomyelitis (branch: deep-tissue complications) →
5. In hosts with impaired defenses (cirrhosis, immunosuppression, neonate), bacteria breach into bloodstream → *results in* bacteremia → hematogenous seeding → meningitis, endocarditis (branch: invasive disease, higher mortality).

Causal chain (veterinary atrophic rhinitis — PMT-driven; mechanistically best characterized)

1. Toxigenic *P. multocida* colonizes the nasal mucosa and secretes **PMT (146 kDa, toxA)** →
2. PMT enters cells and **deamidates a conserved glutamine** in the α -subunit of three of four heterotrimeric G-protein families (Gaq/11, Gai1,2,3, Ga12/13), locking them in the active GTP-bound state (PMID: 23150526) → *results in* constitutive G-protein signaling →
3. Constitutively active **Gaq/11** → **p63RhoGEF** → **RhoA** → **Rho-kinase** → **Ras/MEK/ERK** MAP-kinase cascade (PMID: 23696743) → *leads to* **inhibition of osteoblast differentiation** →
4. In parallel, **Gaq/11** → **PLC β** → **PKC** → **mTORC1** activation → rpS6 phosphorylation, protein synthesis, proliferation (PMID: 23223576; CTGF upregulation contributes, PMID: 23415771) → *drives* remodeling favoring resorption →

5. Net **bone resorption exceeding formation** in the nasal turbinates → *results in* turbinate atrophy and facial distortion (atrophic rhinitis). Prolonged signaling also causes selective membrane redistribution/depletion of Gαq ([PMID: 27490568](#)).

Mechanistic detail

- **Molecular pathways:** Gq/11–RhoA–MAPK (Ras/MEK/ERK); Gq/11–PLCβ–PKC–mTORC1; complement cascade evasion; innate TLR4/LPS signaling.
- **Cellular processes:** Inhibition of osteoblast differentiation, stimulated proliferation, inflammation, phagocytosis evasion.
- **Protein dysfunction:** PMT is a **gain-of-function** enzymatic toxin producing constitutive Gα activation via deamidation. Capsule and LPS are structural virulence surfaces.
- **Immune involvement:** Anti-phagocytic capsule; complement resistance; LPS-driven endotoxic inflammation. Not autoimmune.
- **Tissue damage:** Direct inflammatory tissue destruction and, in atrophic rhinitis, dysregulated bone remodeling.

Suggested ontology terms. GO:0006954 (inflammatory response); GO:0007186 (GPCR signaling); GO:0000165 (MAPK cascade); GO:0038202 (TORC1 signaling); GO:0001503 (ossification, inhibited); GO:0006909 (phagocytosis, evaded). CL:0000062 (osteoblast); CL:0000092 (osteoclast); CL:0000775 (neutrophil); CL:0000235 (macrophage).

Section 7 — Anatomical Structures Affected

- **Primary organs / sites (human):** Skin and subcutaneous soft tissue (UBERON:0002097 skin; UBERON:0002384 connective tissue), especially the **hand and fingers** after cat bites. Lungs/ lower respiratory tract (UBERON:0002048) — second most common site.
- **Secondary / invasive involvement:** Bloodstream (bacteremia), meninges (UBERON:0002360; meningitis), heart valves (endocarditis), joints (UBERON:0000982; septic arthritis), bone (UBERON:0002481; osteomyelitis), liver/biliary tract in cirrhotic sepsis.
- **Body systems:** Integumentary, musculoskeletal, respiratory, cardiovascular, nervous (CNS in meningitis).
- **Tissue/cell level:** Epithelial and connective tissue; osteoblasts (CL:0000062) and osteoclasts (CL:0000092) in atrophic rhinitis; neutrophils and macrophages in acute inflammation.

- **Subcellular:** Host **plasma membrane** (GO:0005886) — site of Gaq depletion/redistribution by PMT; cytosol.
 - **Localization / lateralization:** Typically **unilateral/local** at the bite site; respiratory and invasive forms are systemic.
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Section 8 — Temporal Development

- **Onset:** Local wound infection is **acute and rapid** — erythema, warmth, tenderness and drainage within hours of a bite (PMID: 6371440). Respiratory infection is more **indolent/subacute** (PMID: 11242756). Invasive sepsis can be **fulminant**, with death within hours in vulnerable hosts (PMID: 7560209).
 - **Age of onset:** Any age — neonatal (including vertical transmission) through geriatric.
 - **Progression / course:** With prompt appropriate antibiotics, most local infections resolve; untreated or deep infection progresses to septic arthritis/osteomyelitis. Invasive disease can progress rapidly to death.
 - **Duration:** Generally **self-limited to short-course** with treatment; complicated bone/joint infection may require prolonged therapy.
 - **Remission:** Treatment-induced with antibiotics ± surgical drainage/debridement.
 - **Critical period:** The **first ~8 hours** post-bite is the key window for irrigation, debridement, and prophylaxis decisions (PMID: 39462695).
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Section 9 — Inheritance and Population

Inheritance. Not applicable — infectious, non-heritable. No inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency.

Epidemiology. Human pasteurellosis is sporadic and, in severe/invasive form, uncommon; precise population incidence/prevalence figures are not well established because most data are case series. Bite-wound colonization is common: *Pasteurella* is isolated from ~50% of infected dog bites and ~75% of infected cat bites (PMID: 9887159).

Demographics. Severe disease clusters in neonates/infants, the elderly, and immunocompromised/cirrhotic patients. In one adult cohort (n=44), mean age was 64 years with a female majority (30/44) (PMID: 26356688). Cats are the source in 60–80% of cases (PMID: 6371440).

Geographic distribution. Worldwide, following the distribution of domestic cats and dogs. The **veterinary** disease hemorrhagic septicemia (serotype B:2) is highly endemic in **South Asia** (PMID: 36503053).

Section 10 — Diagnostics

- **Microbiology (gold standard):** Culture of *P. multocida* from wound exudate, blood, sputum, CSF, or joint/pleural fluid. Grows on blood and chocolate agar (not MacConkey); oxidase-, catalase-, and indole-positive. Identification by MALDI-TOF or 16S rRNA sequencing.
- **Laboratory / biomarkers:** Nonspecific inflammatory markers (elevated WBC, CRP). Blood cultures essential when systemic disease suspected — bacteremia in 51% of infantile invasive cases (PMID: 40063964).
- **Imaging:** X-ray/CT/MRI for suspected osteomyelitis, septic arthritis, lung abscess, or empyema; CT/echocardiography for endocarditis; lumbar puncture for meningitis.
- **Genetic / omics testing:** Not applicable for host diagnosis. Pathogen genotyping (capsular multiplex PCR for serogroups A/B/D/E/F; LPS genotyping) is used epidemiologically and in veterinary contexts.
- **Clinical criteria & differential diagnosis:** Diagnosis is clinical (bite history + rapid cellulitis) confirmed by culture. Differential: *Streptococcus/Staphylococcus* cellulitis, *Capnocytophaga canimorsus* (asplenic/cirrhotic), *Bartonella henselae* (cat-scratch disease), anaerobic bite-wound flora. A distinguishing feature is the **unusually rapid** onset of cellulitis (hours, not days) after a cat/dog bite.

Section 11 — Outcome / Prognosis

Prognosis depends on host status and syndrome group. Local soft-tissue infection generally resolves with antibiotics. Invasive disease carries meaningful mortality:

Population / syndrome	Key outcomes	Source
Infantile <i>P. multocida</i> meningitis (n=74)	Bacteremia 51%, seizures 20%, mortality 8%	PMID: 40063964
Neonatal pasteurellosis (n=25)	Overall mortality 20%	PMID: 19208671
Adult cohort, non-bite vs bite (n=44)	Non-bite: bacteremia 37% vs 4% (p=0.001); hospitalized 84% vs 44% (p=0.012); all 4 deaths and 7/8 ICU patients non-bite	PMID: 26356688
Cirrhotic biliary sepsis	Fulminant, fatal within hours (Waterhouse–Friderichsen)	PMID: 7560209

Prognostic factors. Absence of a bite (mucosal/respiratory/indirect acquisition), Charlson comorbidity index ≥ 1 , immunocompromise, cirrhosis, and extremes of age predict worse outcomes ([PMID: 26356688](#)). **Complications** include abscess, tenosynovitis, septic arthritis, osteomyelitis, empyema, endocarditis, and meningitis. Recovery is generally complete with early appropriate therapy; deep bone/joint disease may leave functional deficits.

Section 12 — Treatment

- **First-line pharmacotherapy:** **Penicillin** is preferred; *P. multocida* is characteristically susceptible ([PMID: 9097378](#)). **Doxycycline** is a highly effective alternative ([PMID: 9097378](#)). For polymicrobial bite wounds (aerobes + anaerobes in 56%, [PMID: 9887159](#)), **amoxicillin-clavulanate** is standard because it covers *Pasteurella* plus streptococci, staphylococci, and anaerobes.
- Suggested NCIT terms: NCIT:C716 (Penicillin); NCIT:C263 (Doxycycline); NCIT:C61735 (Amoxicillin-Clavulanate).
- **Alternatives:** Third-generation cephalosporins (ceftriaxone) and fluoroquinolones for serious/invasive disease or penicillin allergy. *Pasteurella* is typically resistant to first-generation cephalosporins and clindamycin.
- **Surgical / interventional:** Irrigation, sharp debridement, drainage of abscesses; joint washout for septic arthritis; negative-pressure wound therapy benefits serious limb bite lacerations ([PMID: 26964825](#)).

- **Supportive:** ICU support, source control, vasopressors in septic shock.
 - **Pharmacogenomics / advanced therapeutics (gene/cell/RNA/immunotherapy):** Not applicable.
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Section 13 — Prevention

- **Primary prevention:** Avoid/limit high-risk animal contact (especially for cirrhotic and immunocompromised individuals); hand hygiene after animal contact; prompt wound care.
 - **Bite-wound management (core prevention of infection):** Immediate copious **irrigation and sharp debridement**; primary closure of well-irrigated, sharply debrided wounds within 8 h is not associated with increased infection; **routine antibiotic prophylaxis in low-risk wounds does not lower infection risk**, so prophylaxis should be selective (high-risk: hand/deep puncture, cat bites, immunocompromised hosts) ([PMID: 39462695](#)). Amoxicillin-clavulanate is the standard prophylactic agent when indicated.
 - **Rabies post-exposure prophylaxis** is indicated for all dog bites where the animal's rabies status cannot be determined or the animal cannot be quarantined for 10 days ([PMID: 39462695](#)).
 - **Human vaccine:** None available.
 - **Veterinary immunization (indirect benefit):** Killed and attenuated (gdhA-derivative) and recombinant OmpH vaccines control hemorrhagic septicemia in cattle/buffalo ([PMID: 32547726](#); [PMID: 30483888](#); [PMID: 42758249](#)). LPS-based immunity is serovar-specific ([PMID: 21664074](#)).
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Section 14 — Other Species / Natural Disease

P. multocida is a **major, naturally occurring veterinary pathogen** — arguably its primary disease niche — and pasteurellosis is fundamentally a zoonosis.

Host	NCBI Taxon	Disease	Serotype / notes
Cattle, buffalo	9913 / 89462	Hemorrhagic septicemia — peracute, ~100% mortality within hours	B:2; endemic South Asia (PMID: 36503053)
Chickens, ducks	9031 / 8839	Fowl cholera	LPS-dependent virulence (PMID: 21664074 ; PMID: 33663572)
Swine	9823	Progressive atrophic rhinitis	PMT-driven (toxA)
Rabbits	9986	"Snuffles," pneumonia	—
Cats, dogs	9685 / 9615	Oral commensal; reservoir for human zoonosis	<i>P. multocida</i> ssp. <i>multocida/septica</i> (cats), <i>P. canis</i> (dogs) (PMID: 9887159)

Transmission / zoonotic potential: High. Cats and dogs are the principal source of human infection (cats 60–80%) via bites, scratches, and licking ([PMID: 6371440](#)). **Comparative pathology:** capsule and LPS virulence mechanisms are conserved across host species; the PMT-Gα deamidation pathway acts on conserved mammalian G proteins, underpinning cross-species relevance.

Section 15 — Model Organisms

- **Mouse (intraperitoneal challenge):** The definitive virulence model. Wild-type B:2 *P. multocida* multiplies rapidly in blood, spleen, and liver; whereas an **acapsular *cexA* mutant is efficiently cleared and readily phagocytosed**, proving the capsule is a crucial virulence determinant ([PMID: 10816499](#)). Resource: MGI (host); Cellosaurus for cell lines.
- **Chicken / duck (natural-host challenge):** Demonstrated that severely truncated-LPS mutants are fully attenuated and that outer-core transferases *pcgD/hptE* contribute differentially to virulence ([PMID: 21664074](#); [PMID: 33663572](#); [PMID: 17517879](#)).
- **In vitro cell models:** Swiss 3T3 fibroblasts, osteoblasts, and Gaq/11 knockout MEFs dissect PMT signaling ([PMID: 23696743](#); [PMID: 23223576](#); [PMID: 23415771](#)).
- **Phenotype recapitulation:** Animal challenge models faithfully reproduce bacteremic systemic disease and permit gene-knockout attribution of virulence to specific factors.

Limitation: murine models represent invasive/veterinary disease, not the human bite-wound cellulitis syndrome; no rodent model captures human host-comorbidity risk stratification.

Key Findings (with statistical evidence)

F001 — Pasteurellosis is a zoonosis from cats and dogs, including indirect and vertical transmission

In a review of 74 infants with *P. multocida* meningitis, **61% (42/74)** arose from indirect household animal contact; bacteremia occurred in 51% (38/74), seizures in 20% (15/74), mortality 8% (6/74) ([PMID: 40063964](#)). In 25 neonatal cases, cat/dog exposure was the major risk (non-traumatic 44%, traumatic 8%, vertical 44%); overall mortality 20% ([PMID: 19208671](#)). *"Most cases occurring from indirect household animal contact (42/74; 61%)."*

F002 — PMT deamidates Gα proteins to constitutively activate Gq/11-RhoA-MAPK and mTORC1

PMT (146 kDa) deamidates a conserved glutamine in the α-subunit of three of four G-protein families ([PMID: 23150526](#)). *"The toxin deamidates an essential glutamine residue of the Gα(i2) subunit, leading to constitutive activation of the G protein."* Downstream, *"Gα(q/11) activates RhoA via p63RhoGEF ... Activated RhoA transactivates the MAP kinase cascade via Rho kinase, involving Ras, MEK and ERK, resulting in inhibition of osteoblast differentiation"* ([PMID: 23696743](#)). PMT also activates mTORC1 via Gαq/11/PLCβ/PKC ([PMID: 23223576](#)) and depletes membrane Gαq ([PMID: 27490568](#)).

F003 — B:2 hemorrhagic septicemia causes near-100% mortality in cattle/buffalo

"Haemorrhagic septicaemia (HS) is an acute infection of cattle and buffaloes caused by the B:2 serotype of Pasteurella multocida. This disease is highly endemic in South Asia. In some peracute cases, there is 100% mortality in infected animals within a few hours of infection" ([PMID: 36503053](#)).

F004 — Soft tissue is the commonest human site, respiratory second; penicillin first-line

"The respiratory tract is the second most common site of Pasteurella infection after soft tissue infection. Most patients with Pasteurella pulmonary infection are elderly with underlying lung disease, either COPD, bronchiectasis, or malignancy" and "The preferred drug for the treatment of Pasteurella infections is penicillin. Alternately, doxycycline is highly effective" (PMID: 9097378).

F005 — Capsule and LPS are the key virulence factors

"Key virulence factors identified to date include capsule and lipopolysaccharide. The capsule is clearly involved in bacterial avoidance of phagocytosis and resistance to complement, while complete lipopolysaccharide is critical for bacterial survival in the host" — plus PMT, adhesins, and iron-acquisition proteins (PMID: 17107417).

F006 — Non-bite infection is more invasive and lethal than bite infection

Retrospective cohort (n=44): *"Patients presenting without a bite were more frequently bacteremic (37% vs 4%, $P = 0.001$), and were hospitalized more often (84% vs 44%, $P = 0.012$)."* All 4 deaths and 7/8 ICU patients were non-bite-related; Charlson index ≥ 1 associated with absence of a bite ($p=0.006$) (PMID: 26356688).

F007 — Prevention rests on wound care, selective prophylaxis, and rabies PEP

"While dog bites are associated with an overall high rate of infection (6-25 %), routine antibiotic prophylaxis in low-risk wounds does not lower that risk. Postexposure prophylaxis for rabies is indicated for all dog bites where the rabies status of the dog cannot be determined, or the animal cannot be quarantined for 10 days" (PMID: 39462695).

F008 — Capsule is a proven virulence determinant in a mouse model

The B:2 *cap* locus has 15 genes in three regions; disrupting export gene *cexA* yields an acapsular mutant. *"Following intraperitoneal challenge of mice, the acapsular bacteria were removed efficiently from the blood, spleen, and liver; while wild-type bacteria multiplied rapidly" (PMID: 10816499).*

F009 — *Pasteurella* is the most frequent bite-wound isolate

Prospective multicenter study (18 EDs): *"Pasteurella species were the most frequent isolates from both dog bites (50 percent) and cat bites (75 percent). Pasteurella canis was the most common isolate of dog bites, and Past. multocida subspecies multocida and septica were the most common isolates of cat bites"* (PMID: 9887159). Wounds were polymicrobial (aerobes+anaerobes in 56%).

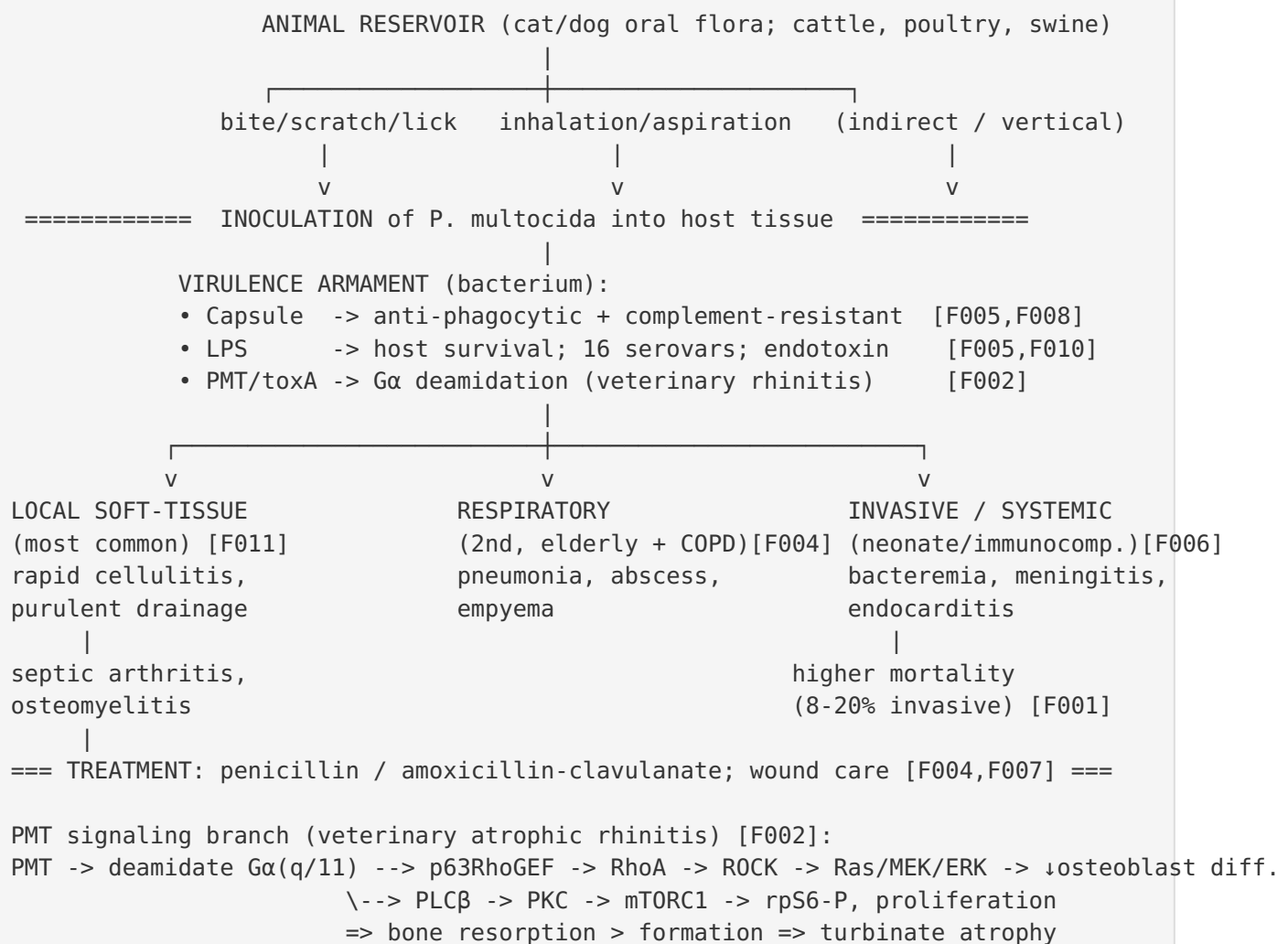
F010 — LPS is essential; basis of 16-serovar classification; serovar-specific immunity

"P. multocida strains are classified into 16 serovars based on lipopolysaccharide (LPS) antigens. LPS is an essential virulence factor of P. multocida; mutants expressing severely truncated LPS are completely attenuated in chickens" and "protection ... is generally considered to be serovar specific" (PMID: 21664074).

F011 — Three human syndrome groups; cats source in 60–80%

"Cats are the source of infection in 60 to 80% of cases and dogs in the great majority of the remainder. Local infections are characterized by the rapid appearance of erythema, warmth, tenderness, and frequently purulent drainage" with serious complications including "septic arthritis proximal to bites or scratches, osteomyelitis" (PMID: 6371440).

Mechanistic Model / Interpretation



The unifying narrative: pasteurellosis is a **bacterial-virulence-driven** disease in which host outcome is a function of (a) inoculation route and (b) host immune competence. The capsule and LPS are the "survival kit" that lets the organism evade phagocytosis and complement and persist long enough to cause disease — proven by the clearance of acapsular and truncated-LPS mutants in animal models. Once established, disease severity bifurcates: an immunocompetent host with a bite develops localized, treatable cellulitis, whereas a compromised host (or a non-bite/mucosal acquisition, which itself flags comorbidity) develops invasive, potentially fatal disease. PMT represents a distinct, mechanistically elegant branch dominant in veterinary atrophic rhinitis, hijacking universal mammalian G-protein signaling.

Evidence Base

PMID	Title (abbrev.)	Supports
6371440	<i>Pasteurella multocida</i> infections. Report of 34 cases	Three-group clinical framework; cats 60–80%; rapid cellulitis (F011)
9097378	<i>Pasteurella multocida</i> pneumonia	Soft tissue > respiratory; penicillin/doxycycline (F004)
26356688	Clinical Features and Outcomes of <i>P. multocida</i>	Non-bite = more bacteremia/ICU/death (F006)
40063964	Infantile <i>P. multocida</i> Meningitis	Indirect transmission 61%; invasive outcomes (F001)
19208671	Neonatal pasteurellosis review	Vertical transmission; 20% mortality (F001)
17107417	<i>P. multocida</i> pathogenesis: 125 years after Pasteur	Capsule + LPS key virulence factors (F005)
10816499	Capsule is a virulence determinant	Acapsular mutant cleared in mice (F008)
21664074	<i>P. multocida</i> LPS: the long and short of it	LPS essential; 16 serovars; serovar-specific immunity (F010)
9887159	Bacteriologic analysis of dog/cat bites	<i>Pasteurella</i> dominant isolate (F009)
23150526	Substrate specificity of PMT for Ga	PMT deamidation mechanism (F002)
23696743	PMT prevents osteoblast differentiation	Gq/11–RhoA–MAPK axis (F002)
23223576	mTORC1 in PMT signaling	PLC β /PKC–mTORC1 branch (F002)
36503053	ELISA diagnostics for HS	B:2 hemorrhagic septicemia, ~100% mortality (F003)
39462695	Surgical Management of Pediatric Dog Bites	Wound care, selective prophylaxis, rabies PEP (F007)
26964825	NPWT for serious dog bites	Adjunctive surgical management
7560209	Waterhouse-Friderichsen in cirrhosis	Fulminant sepsis in cirrhotic host
11242756	Cat cuddler's cough	Inhalation/aspiration respiratory route

Evidence source types: Human clinical (case series, retrospective cohorts): F001, F004, F006, F007, F009, F011. Model organism / in vitro (mouse, chicken/duck, cell lines): F002, F003, F005, F008, F010. This mix means mechanistic virulence claims are strongly supported experimentally, while human epidemiologic precision is limited to observational series.

Limitations and Knowledge Gaps

1. **No population-level incidence/prevalence.** Human data are case series and single-center cohorts; true incidence per 100,000 is unknown. Reported outcome percentages come from small samples (n=25–74).
 2. **Selection/publication bias** toward severe and unusual cases (meningitis, sepsis, fatalities) likely overstates severity relative to the many mild, culture-unconfirmed cellulitis cases treated empirically in primary care.
 3. **Mechanistic data are mostly veterinary/model-organism.** The PMT Gα-deamidation pathway and capsule/LPS virulence are best characterized in atrophic rhinitis, fowl cholera, and mouse models — their direct role in **human** soft-tissue and invasive disease is inferred, not demonstrated.
 4. **No human host-genetic susceptibility data.** Whether host polymorphisms (complement, TLR4, mannose-binding lectin) modulate risk is unstudied.
 5. **Antibiotic resistance trends** in human isolates are sparsely characterized; most susceptibility data are veterinary ([PMID: 24612952](#)).
 6. **Diagnostics lag:** no rapid point-of-care test; culture and MALDI-TOF remain the mainstay, delaying species/serotype confirmation.
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Proposed Follow-up Experiments / Actions

1. **Epidemiologic quantification:** Mine a large EHR/administrative dataset (ICD-10 A28.0) to estimate true human incidence, syndrome distribution, and case-fatality with confidence intervals — filling the F001/F011 quantitative gap.
2. **Host-susceptibility study:** Case-control genotyping (complement, TLR4, MBL2) in invasive vs local pasteurellosis to test whether host innate-immune variants explain the comorbidity-independent severity of non-bite disease (F006).

3. **Mechanistic bridge to human tissue:** Test capsule/LPS mutants (F008/F010) and PMT signaling (F002) in human keratinocyte/osteoblast and whole-blood models to confirm animal-derived virulence mechanisms operate in human infection.
 4. **Antibiotic surveillance:** Systematic collection of human *P. multocida* isolates for β -lactamase and MIC profiling to validate continued reliance on penicillin/amoxicillin-clavulanate (F004).
 5. **Prophylaxis RCT:** A stratified randomized trial of selective antibiotic prophylaxis in cat-bite (high-risk) vs dog-bite (variable-risk) wounds to refine F007's "selective prophylaxis" recommendation.
 6. **Rapid diagnostics:** Develop/validate a multiplex PCR (capsular serogroup + species) directly from wound swab and blood to shorten time to targeted therapy.
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Consensus Answer

Pasteurellosis is a zoonotic bacterial infection caused by *Pasteurella* species (chiefly *P. multocida*), acquired from the oral/respiratory flora of cats and dogs (cats the source in 60–80% of human cases) via bites, scratches, licking, or inhalation. It presents in three groups — rapidly progressive soft-tissue cellulitis at the wound (most common), respiratory infection in elderly patients with chronic lung disease, and invasive disease (bacteremia, meningitis, endocarditis) concentrated in neonates and immunocompromised/cirrhotic hosts — and has no human genetic etiology; pathogenesis is driven by bacterial virulence factors (capsule and LPS for immune evasion, and *Pasteurella multocida* toxin, which deamidates G α proteins to activate Gq/11–RhoA–MAPK/mTORC1 signaling in veterinary atrophic rhinitis). It is treated first-line with penicillin or amoxicillin-clavulanate, and prevention rests on bite-wound care, selective prophylaxis, rabies post-exposure prophylaxis, and hygiene after animal contact.
